Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

MQTT PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A IoT & Edge

TOTAL: 10 QUESTIONS

Q1 What is MQTT and what problem in IoT & Edge does it solve? Threat Model

Q2 What are the core components or concepts in MQTT? Architecture

Q3 How does MQTT compare to its main alternatives? Configuration

Q4 How do you get started with MQTT for a new project? Hardening

Q5 What configuration options most affect MQTT's behavior in production? SSO / IdP

Q6 How does MQTT handle reliability and high availability? Observability

Q7 What observability does MQTT provide out of the box? Lifecycle

Q8 What are common performance bottlenecks in MQTT? Compliance

Q9 What security best practices apply when running MQTT? Testing

Q10 What mistakes do teams commonly make when adopting MQTT? Tuning

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 **MQTT is a tool or platform that** — Mitigation

MQTT is a tool or platform that automates or simplifies a specific concern in the IoT & Edge domain, reducing manual effort and operational complexity for engineering teams.

A2 **MQTT has key building blocks that work** — Primitives

MQTT has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 **MQTT trades off simplicity against feature richness** — Hardening

MQTT trades off simplicity against feature richness differently than its alternatives. Choose MQTT when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 **Install MQTT via its package manager or** — Gotchas

Install MQTT via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 **Production-critical settings typically include concurrency** — Integration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 **MQTT provides HA through leader election, replication,** — Observability

MQTT provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 **MQTT exposes metrics (Prometheus endpoint or built-in** — Automation

MQTT exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 **Performance bottlenecks typically stem from misconfigured** — Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 **Run MQTT with the principle of least** — Assessment

Run MQTT with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 **Common pitfalls include skipping the documentation and** — Containerization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.